

Frame transport for framed formal monodromy:
the lemma, its hypotheses, and the receiver problem
Fifth version, closing the coalesced case

Memo for a reader of *Irrationality of Cubic Threefolds after One Stabilization*

15 August 2026

Abstract

Proposition 4.7 of the draft transports framed formal monodromy across the comparison isomorphisms of Iritani and Iritani–Koto, and the transport step itself is asserted in one sentence. This memo states and proves the missing lemma over a differential field, with its hypotheses made explicit; shows that those hypotheses are jointly unsatisfiable over the coefficient receiver the draft builds, so that the lemma as proved has no verified instance in the application; corrects the receiver; and reduces the remaining question to one inequality that depends on a normalization the draft is free to choose. The inequality fails under the draft’s current instruction to take the separating weight parameter large, and holds at the cubic endpoint under the smallest admissible choice. the archived iterated-receiver discussion examines an iterated receiver that would satisfy the inequality trivially, checks it against the sources, and finds that it needs a semipositivity hypothesis on the ambient variety which weak factorization does not supply.

The second version of this memo proposed the framing theory of Hinault, Yu, Zhang and Zhang as the probable repair, on the ground that its decomposition gauge is regular in the loop coordinate. Three compatibility checks against the pinned sources have since been run, and the route does not survive them; Section 4 now reports what those checks found. Two consequences are worth more than the route was. First, no receiver is needed for the comparison isomorphisms at all: Iritani’s Ψ and Iritani–Koto’s Φ are polynomial in the loop coordinate with inverses in the same ring, so the transport theorem below applies to them with $\Gamma = 0$. Second, the diagnosis of the second version was wrong — the pro-Laurent bulk gauge is not the draft’s own choice but Iritani–Koto’s own fundamental solution, which the sources meet and handle by Birkhoff factorization. What is left is a single statement about one bulk displacement, recorded in Section 4. Section 5 then attacks that statement directly through the flatness identities. It proves that multiplicity-one blocks carry no framed monodromy, which is unconditional and leaves only coalesced blocks in play, and writes down the exact evolution of the block’s leading operator. It does *not* reduce the coalesced case: the third version of this memo claimed such a reduction and was wrong twice over, once in dropping a term from the off-block Sylvester equation and once in modelling a nonsemisimple block by its compressed pair. Both errors, and the cubic 2×2 example that is the regression test for any repair, are recorded there.

Section 7, new in this version, closes the coalesced case for a block of the cubic’s type, and with it the residual gap for the cubic ambient summand. The device is to shear the block before deforming it. Three statements are proved over the formal even bulk germ, from the flatness identities and nothing else: the double eigenvalue never splits, because $\det N$ satisfies $\partial_a \det N = 2q_a \det N$ and vanishes initially; no irregularity appears, because the commutant of a regular block forbids a z^{-2} bulk term and the z^{-1} flatness equation then forces the surviving pole to vanish; and the residue of the sheared block moves by conjugation alone, so

its characteristic polynomial is constant. For the cubic that polynomial is $\rho^2 + \rho + 5/36$ with roots $-1/6, -5/6$, so the primitive-sixth multiplicity is 2 at every bulk parameter, including a specialized one. No receiver, no bulk gauge, and no convergence question enters, because specialization is substitution into constants.

Section 8 assembles the one-stabilization statement — that $X \times \mathbf{P}^1$ is irrational for every smooth cubic threefold — in the atom formulation, where the intermediate varieties of a weak factorization cancel in the ledger instead of being visited, so that no framed operator is transported across a comparison isomorphism and Hypothesis 4.7H does not arise. Every step is proved there except one, and that one is stated as the gap it is: the primitive-sixth count must be invariant under change of base point within a connected component of the spectral cover, which is part of the definition of an atom. Cai’s gauge argument gives that invariance along a formal germ, and it does not globalize, because the gauge is the pro-Laurent object whose sum over bulk degrees loses its lower bound in the loop coordinate outside topologically nilpotent parameters. So the receiver difficulty of the earlier sections reappears at the globalization step rather than being escaped. Two consequences are recorded: the low-dimensional side needs no such invariance, because the sources’ nef-canonical claim is pointwise structural; and the sources’ own architecture avoids the gap by decorating with a Serre automorphism, whose well-definedness on atoms follows from rigidity of representations of a proreductive group rather than from any constancy of monodromy. Section 9 carries the endpoint most of the way. An earlier version of it claimed the theorem outright by inverting the comparison’s change of variables; that was refuted, since the sources’ invertibility is in displaced coordinates treated as formal variables and the substitution the argument needed is ill-defined. What survives the refutation is stronger than what motivated it. For the trivial rank-two bundle the displacement is $\varsigma_j^\circ = r\lambda_j = \pm 2q^{1/2}$ *exactly*, a pure string-direction shift with no tail — proved twice, once from the sources’ own formula and once by a spectral argument on the product that needs no transform at all. An H^0 shift changes no exponent, so $\nu_6(X \times \mathbf{P}^1, 0) = 4$ is a computed fact rather than a hypothesis, and the standing hypothesis is discharged at the endpoint rather than relocated. Arranging the weak factorization as a roof then makes every substitution a forward one, so no inverse is evaluated anywhere. What is left is a single statement about a single variety: that $X \times \mathbf{P}^1$ keeps at least one primitive-sixth eigenvalue along the accumulated displacement. Its ν_6 -carrying blocks are two copies of the cubic’s rank-two Jordan block, which are exactly the hypotheses of Section 7, and the pencil along which they must be followed is polynomial per slot, so the engine to run there is Section 7 rather than a bulk gauge. Section 12 lists what the earlier versions got wrong; readers of those versions should start there.

1 The assertion at issue

References are to Section 4 of the draft, *Framed formal monodromy*. I name statements by their semantic labels as well as by number, because the edits proposed in Section 11 renumber the section: Definition 4.1 (`def:framed-sixth-multiplicity`), Definition 4.2 (`def:pro-laurent-gauge-group`), Lemma 4.3 (`lem:pv-base-change`), Lemma 4.5 (`lem:formal-base-shift`), Proposition 4.7 (`prop:framed-operations`), Lemma 4.9 (`lem:divisor-tagging`).

Lemma 4.3 ends with the sentence

If the gauge uses only integral powers of z at every finite level, this conjugacy preserves the original- z -disc frame,

whose proof is the single sentence “the deck action fixes every integral power of z at each finite

level, so the compatible gauge is single-valued for the framed turn.” Lemma 4.5 then *defines*

$$M^{\text{bulk}} = G M_{\mathfrak{f},V}^{\text{small,shifted}} G^{-1} \quad (4.1c)$$

and Proposition 4.7 concludes that “each center target summand has the framed monodromy of its specialized small connection.” The conclusion does not follow from the display: conjugation preserves a characteristic polynomial for trivial reasons, but says nothing about whether the conjugated matrix is the intrinsic turn operator on horizontal solutions of the gauged module. What is missing is a transport lemma together with an algebra in which the turn acts on solutions.

2 The lemma over a differential field

Notation for Exp_V , $K = \mathbf{C} \oplus V$ and Ω_V is the draft’s, from (4.0) and (4.0a). Recall that Exp_V is defined on $(K, +)$, not on Ω_V .

The ground field

Let Γ be a totally ordered abelian group and let

$$\mathcal{H} = \Omega_V((\Gamma \oplus \mathbf{Z}e_z))$$

be the Hahn field of series with well-ordered support and coefficients in Ω_V , the value group ordered lexicographically with the e_z -coordinate *last*, as in the draft. Write $z = x^{e_z}$ and let θ multiply the coefficient at (γ, k) by k ; this is a derivation because k is additive on the value group, and $\theta z = z$. Its constants are

$$\mathcal{C} := \mathcal{H}^{\theta=0} = \Omega_V((\Gamma)).$$

Taking $\Gamma = 0$ gives $\mathcal{H} = \Omega_V((z))$. For $e \geq 1$ put $\mathcal{H}_e = \Omega_V((\Gamma \oplus \frac{1}{e}\mathbf{Z}e_z))$, a degree- e extension containing $w = x^{e_z/e}$, and $\theta_w = e\theta$, so $\theta_w w = w$. All residues below are taken with respect to θ_w on the ramified side; the draft’s $M_{\text{RS},V}$ uses the same convention, and the two differ by the factor e if one is careless.

The solution algebra and the turn

Definition 1. Assume the exponential factors under consideration have *constant* coefficients, that is $\varphi \in w^{-1}\Omega_V[w^{-1}]$. The *universal formal solution algebra* $\mathcal{U}_e$ is the $\mathcal{H}_e$ -algebra generated by symbols

$$w^\rho \ (\rho \in K), \quad \ell, \quad E(\varphi),$$

subject to $w^\rho w^{\rho'} = w^{\rho+\rho'}$, w^n = the element w^n of $\mathcal{H}_e$ for $n \in \mathbf{Z}$, $E(\varphi)E(\varphi') = E(\varphi + \varphi')$, $E(0) = 1$, with

$$\theta_w(w^\rho) = \rho w^\rho, \quad \theta_w(\ell) = 1, \quad \theta_w(E(\varphi)) = \theta_w(\varphi)E(\varphi).$$

The two hypotheses buried in that definition — constant coefficients in φ , and exponents ρ in K rather than in Ω_V — are not bookkeeping. The first is false for the modules the draft transports and is discussed at Remark 5; the second is forced because Exp_V is defined only on K , so a module whose residues leave K needs Exp rebuilt over a larger field.

Lemma 2 (Structure). *Fix representatives $\Lambda \subset K$ for $K/\mathbf{Z}$ with $0 \in \Lambda$. Then $\mathcal{U}_e$ is free as an $\mathcal{H}_e$ -module with basis $\{w^\rho \ell^m E(\varphi)\}$, $\rho \in \Lambda$, $m \geq 0$. In particular $\mathcal{U}_e \neq 0$ and $\mathcal{H} \subset \mathcal{U}_e$.*

Proof. One cannot argue that $K/\mathbf{Z}$ is torsion-free; it is not, since it contains $\mathbf{Q}/\mathbf{Z}$, and the rational classes are exactly the ones ν_6 counts. Instead use base change. The group algebra $\mathcal{H}_e[K]$ is free over $\mathcal{H}_e[\mathbf{Z}] = \mathcal{H}_e[t, t^{-1}]$ on any set Λ of coset representatives, and $\mathcal{U}_e$ is obtained from it by the base change $\mathcal{H}_e[t, t^{-1}] \rightarrow \mathcal{H}_e$, $t \mapsto w$, a surjection with kernel $(t - w)$ generated by a nonzerodivisor. Base change of a free module is free on the image of the same basis. The crossed multiplication $w^\rho w^{\rho'} = w^n w^{\rho''}$ is well defined, and the derivation is consistent with the integer identification because w^n sits at e_z -exponent n/e and θ_w multiplies it by n . The factors $E(\varphi)$ form the group algebra of a torsion-free group and ℓ is a polynomial variable. $\square$

Lemma 3 (The turn). *There is a unique automorphism σ of $\mathcal{U}_e$ with*

$$\sigma(w) = e^{2\pi i/e} w, \quad \sigma(w^\rho) = \text{Exp}_V(\rho/e) w^\rho, \quad \sigma(\ell) = \ell + \frac{2\pi i}{e}, \quad \sigma(E(\varphi)) = E(\sigma\varphi),$$

acting on $\mathcal{H}_e$ by multiplying the coefficient at $(\gamma, j/e)$ by $e^{2\pi i j/e}$. It fixes $\mathcal{H}$, hence $\mathcal{C}$, pointwise, and commutes with θ_w .

Proof. The action on $\mathcal{H}_e$ multiplies coefficients by a character of the value group, hence is a support-preserving ring automorphism, and is the identity on $\mathcal{H}$ because there $j \in e\mathbf{Z}$. Multiplicativity in ρ is the homomorphism property of Exp_V on $(K, +)$, and the two prescriptions agree on integral exponents because $\text{Exp}_V(n/e) = e^{2\pi i n/e}$ by (4.0a). Commutation with θ_w is not a diagonality argument, since neither $\sigma(\ell) = \ell + 2\pi i/e$ nor $\theta_w(\ell^m) = m\ell^{m-1}$ is diagonal; it is the direct check $\sigma\theta_w(\ell^m) = m(\ell + \frac{2\pi i}{e})^{m-1} = \theta_w\sigma(\ell^m)$, together with the diagonal cases. $\square$

Lemma 4 (Constants). *Under the constant-coefficient hypothesis of Definition 1, $\mathcal{U}_e^{\theta_w=0} = \mathcal{C}$.*

Proof. Let v be the Hahn valuation of $\mathcal{H}_e$, additive because the coefficient field is a field; θ_w multiplies coefficients by integers and never enlarges a support, so $v(\theta_w c) \geq v(c)$.

Write $x = \sum c_{\rho, m, \varphi} w^\rho \ell^m E(\varphi)$ in the basis of Lemma 2 with $\theta_w x = 0$; the equation splits over φ . For $\varphi \neq 0$, let M be the top ℓ -degree in that block; for each ρ , $\theta_w(c_{\rho, M}) + \rho c_{\rho, M} + \theta_w(\varphi) c_{\rho, M} = 0$. Every monomial of $\theta_w(\varphi)$ has Γ -part 0 and strictly negative e_z -part, so $v(\theta_w(\varphi)c) < v(c) \leq v(\theta_w c + \rho c)$ for $c \neq 0$ and the lowest term cannot cancel; hence $c_{\rho, M} = 0$, and descending in M kills the block.

For $\varphi = 0$ the coefficient equations are $\theta_w(c_{\rho, m}) + \rho c_{\rho, m} + (m+1)c_{\rho, m+1} = 0$. Descending in m : the top-degree coefficient satisfies the homogeneous equation $\theta_w(c) + \rho c = 0$, and on each monomial this reads $(j + \rho)c = 0$ with $j \in \mathbf{Z}$, so $c = 0$ unless $\rho \in \mathbf{Z}$, that is unless $\rho = 0$. For $\rho = 0$ the top coefficient is θ_w -constant and the next equation gives $M c_{0, M} = -\theta_w(c_{0, M-1})$, whose right side has zero component in e_z -degree 0; comparing that component gives $c_{0, M} = 0$ for $M \geq 1$. What survives is $\rho = 0$, $m = 0$, $\varphi = 0$ with θ_w -constant coefficient, and the θ_w -constants of $\mathcal{H}_e$ are those supported in e_z -degree zero, namely $\mathcal{C}$. $\square$

Remark 5 (The constant-coefficient hypothesis is necessary, and fails in the application). Allow φ a Novikov coefficient of positive weight, $\varphi = x^\gamma w^{-1}$ with $\gamma \in \Gamma$, $\gamma > 0$ — the shape of every exponential factor the draft actually meets, since the eigenvalues λ of $c_1 \star$ have positive Novikov weight. Then $v(\theta_w \varphi) > 0$ in the coefficient-dominant order and the domination argument reverses. Worse, the equation $\theta_w(c) = \theta_w(\varphi)c$ is solved by

$$c = \sum_{n \geq 0} \frac{(-1)^n}{n!} x^{n\gamma} w^{-n},$$

whose support $\{(n\gamma, -n/e)\}_{n \geq 0}$ is increasing, hence well ordered, so $c \in \mathcal{H}_e$ and $cE(\varphi)$ is a θ_w -constant outside $\mathcal{C}$. Lemma 4 is therefore false in that regime, and with it Proposition 6. The structural reason is worth stating on its own: *in the draft's coefficient-dominant order, $\exp(-\lambda/z)$ is already a unit of the coefficient field*, so adjoining $E(\varphi)$ as a free symbol duplicates an existing element and inflates the constants. In that order the exponential factors are not symbols at all, and the entire irregularity of the formal solution sits in its divergent unit part, which is exactly where the rate comparison of Section 3 lives.

Framed operators and transport

Proposition 6 (The framed operator is the matrix of the turn). *Let D be a differential module over $\mathcal{H}$, free of rank n , with connection matrix Ω , and let $Y \in \mathrm{GL}_n(\mathcal{U}_e)$ satisfy $\theta Y = -\Omega Y$. Then $\mathrm{Sol}(D) = \{v : \theta v + \Omega v = 0\} = Y\mathcal{C}^n$, free of rank n and σ -stable; the matrix M with $\sigma(Y) = YM$ lies in $\mathrm{GL}_n(\mathcal{C})$ and is well defined up to $\mathrm{GL}_n(\mathcal{C})$ -conjugacy. For D over $\Omega_V((z))$ in Levelt–Turrittin form, M is the draft's framed operator: in a solution basis adapted to one deck orbit of length d the matrix of σ is block-cyclic with the draft's intertwiners A_i as blocks, and the cyclic-block determinant gives $\chi(X) = \det(X^d I - U)$ with $U = A_{d-1} \cdots A_0$, which is (4.0b).*

Proof. From $\theta(Y^{-1}) = Y^{-1}\Omega$ one gets $\theta(Y^{-1}v) = Y^{-1}(\theta v + \Omega v)$, so horizontality of v means $Y^{-1}v$ has θ_w -constant entries, and Lemma 4 identifies those with $\mathcal{C}$. Since σ commutes with θ and fixes Ω , it preserves $\mathrm{Sol}(D)$; so $M = Y^{-1}\sigma(Y)$ has entries in $\mathcal{C}$ and is invertible, being a product of invertibles, with inverse $\sigma(Y^{-1})Y$. Two solution matrices differ on the right by an element of $\mathrm{GL}_n(\mathcal{C})$, which conjugates M . $\square$

Theorem 7 (Frame transport). *Let D, D' be as in Proposition 6, with solution matrices Y, Y' , and let $G \in \mathrm{GL}_n(\mathcal{H})$ satisfy $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$. Then GY is a solution matrix for D' and $\sigma(GY) = (GY)M$. Consequently*

$$M_{f,V}(D') = C_0^{-1} M_{f,V}(D) C_0 \quad \text{for some } C_0 \in \mathrm{GL}_n(\mathcal{C}),$$

so the two framed operators have the same characteristic polynomial in $\mathcal{C}[X]$ and the same multiplicity at every root of unity.

Proof. $\theta(GY) = \theta(G)Y - G\Omega Y = -\Omega'(GY)$. Since σ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$, so $\sigma(GY) = G\sigma(Y) = GYM$. Writing $Y' = (GY)C_0$ with $C_0 \in \mathrm{GL}_n(\mathcal{C})$ gives $M' = C_0^{-1}MC_0$. $\square$

Remark 8 (Which object is $M_{f,V}$). The framed operator is an automorphism of the horizontal module; its matrix depends on a choice of *solution* frame and is well defined up to conjugacy over the constants. The conjugate GMG^{-1} appearing in the draft's (4.1c) is not the matrix of the turn in any solution frame — Y' differs from GY by a constant matrix, not by G — but the matrix of the transported automorphism read in the *module* frame of D' . Both readings are legitimate, and they give the same characteristic polynomial, but the draft should say which it means, since conflating them is exactly the step at issue.

Remark 9 (The hypothesis on G is the whole content). Since σ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$ is automatic once G has entries in $\mathcal{H}$; what matters is that G lies in $\mathcal{H}$ and not in $\mathcal{U}_e$. No hypothesis-free statement exists: Y itself is a gauge over $\mathcal{U}_e$ carrying D to the trivial module, whose framed operator is the identity. The rank-one case shows what survives: if $g \in \mathcal{H}^\times$ and $\theta(g)/g = \delta$, then each monomial gives $(k - \delta)g_{(\gamma,k)} = 0$, so $\delta \in \mathbf{Z}$; a gauge over $\mathcal{H}$ shifts residues only by integers, which Exp_V kills.

3 The receiver

The draft's receiver, in the proof of Proposition 4.7, is

$$\mathcal{R}_j = \Omega_{V,j} \otimes_{H_{0,j}} H_{\text{tot},j}, \quad H_{\text{tot},j} = \mathbf{C}((\Gamma_j^{\text{coeff}} \oplus \mathbf{Z}e_z)). \quad (4.4c)$$

The two rates

Weight a monomial of coefficient weight d and z -power k by $a d + b k$, $a, b > 0$, and let ε be the minimal weight of a generator of J_j .

- *Pro-Laurent gauges gain one filtration unit per unit of z -pole.* By the recursion (4.1a) and the draft's own conclusion, G_α has no power of z below $-|\alpha|$ and a term of bulk degree m lies in $F^m B$; every monomial of J_j^N has weight at least N . Their weights grow like $m(a-b)$, which together with the draft's finite-below argument gives well ordered support when $a > b$.
- *Splitting gauges lose $w(\Delta\lambda)$ per unit of z -power.* This is the draft's own construction (4.9d): at order n the block off-diagonal coefficient is removed by a Sylvester equation $[K_0, A_n] = B_n$ with K_0 the leading matrix, so one division by an eigenvalue difference per order, and $w(A_n) \geq -n w(\Delta\lambda)$. The draft's displayed cubic entry $-14/(81r^2)$ at z -order two exhibits the rate: one factor of r^{-1} per unit of z -power. Three caveats. After a ramification $z = w^e$ the divisions occur once per w -order, so the loss is $e w(\Delta\lambda)$ per unit of z -power; at the cubic the draft proves the exponential blocks are unramified in z , so $e = 1$ there. The claim that the regular-singular normalization costs nothing needs the homogeneity argument that residue eigenvalues are rational, verified in the cubic block where $\det L_s = (s + 1/6)(s + 5/6)$. And $w(\Delta\lambda)$ is not controlled by $w(\lambda)$ when the Picard rank exceeds one, since w and the quantum grading are different functionals and leading terms can cancel.

Both conditions hold together iff

$$e w(\Delta\lambda) < \varepsilon$$

The criterion is not an artifact of choosing an order

For $C := e w(\Delta\lambda)/\varepsilon \geq 1$ the obstruction is not about well ordering at all. The z^n -coefficient of the single product GP of a pro-Laurent gauge with a splitting gauge is $\sum_m g_m p_{n+m}$, whose terms have weight at least $m(1-C) - nC$; for $C \geq 1$ infinitely many terms have bounded weight, so that coefficient is an infinite sum in no completion, ordered or not. Any repair that forms this product must therefore establish $C < 1$. A repair that never forms it — see route 2 below — is not bound by this.

The criterion depends on a normalization the draft may choose

This is the practical point, and it reverses the recommendation of the first version of this memo. The draft fixes $w(u) = 1$ and $w(s_{j,\ell}) = 1$ independently of L , and sets $w(Q^{i \cdot d} u^{\rho_C \cdot d}) =$

$L(H \cdot i_* d) + \rho_C \cdot d$. Hence $\varepsilon = 1$ always, while the weights of the Novikov generators grow linearly in L . The criterion is therefore

$$e(L(H \cdot i_* d_0) + \rho_C \cdot d_0) < c_1 \cdot d_0,$$

using that the eigenvalues of $c_1 \star$ are homogeneous of quantum degree one when Q^d has degree $c_1 \cdot d$ — a convention I have checked against the draft’s own low-dimensional vanishing proof, where the Q^β -component of $c_1 \star$ maps H^i to $H^{i+2-2c_1 \cdot \beta}$, and against its explicit cubic block, where K_0 has eigenvalues $\pm 6r, 0, 0$ with $r = (3q)^{1/2}$ and $c_1 \cdot \ell = 2$. Consequences:

- Under the draft’s instruction to choose L “so large that” separation holds, the criterion fails at essentially every comparison with two distinct exponential factors — including the cubic endpoint.
- Under the smallest admissible L it can hold. For a projective bundle with $c_1(V) \cdot \ell = 0$ one may take $L = 1$, giving $w(Q^\ell) = 1 = \varepsilon$ and $w(\Delta\lambda) = w(q^{1/2}) = 1/2$.
- Independently of L : separation forces $w \geq \varepsilon$ on every generator of J_j , and the image of Q^{d_0} is a generator, so $c_1 \cdot d_0 \geq 2$ is necessary under any admissible weight. Index one fails outright.

So the cheapest available repair is to choose the separating weight minimally rather than large, and to check the resulting inequality case by case. The first version of this memo asserted the opposite, on the false ground that the ratio was scale-invariant; rescaling w does move both rates equally, but L is not a rescaling, because the bulk generators’ weights are pinned at one.

4 The comparison maps, and the residual gap

The framing route proposed in earlier versions — reconstructing the framed operator through the Hinault–Yu–Zhang–Zhang F -bundle formalism — was checked against the pinned sources and closed. The route as proposed and the three checks are archived; their verdicts, locators and hashes are recorded in `2026-08-15-c912-framing-compatibility-checks.md`. Check one fails, check two passes, and check three confirms that the sources cannot certify an intrinsic centre invariant. What survives from that exercise is stated next, and it is worth more than the route was.

What survives, and the corrected diagnosis

Two things survive, and both are worth more than the route was.

The comparison isomorphisms need no receiver. Iritani’s Ψ is an isomorphism of $\mathbf{C}[z]((q^{-1/s}))[[Q, \hat{\tau}]]$ -modules (Theorem 5.18), and Iritani–Koto’s Φ is induced by a $\mathbf{C}[z, q][[Q, \hat{\tau}]]$ -module map and is an isomorphism over $\mathbf{C}[z]((q^{-1/r'}))[[Q, \hat{\tau}]]$, with $r' \in \{r, 2r\}$ according to the parity convention of their statement (proof of their Theorem 5.1(6)). An isomorphism of modules has a module-map inverse, so Ψ^{-1} has entries in the same ring with no separate argument. Both carry only *non-negative integral* powers of z — power series in z , not polynomials in z , since Iritani’s Remark 1.5 identifies the ring with $\mathbf{C}[q^{\pm 1/s}][[Q, \hat{\tau}]][[z]]$; an earlier version of this memo said “polynomial”, which is too strong, while the draft’s own “integral powers of z ” is right. What matters is that no negative power of z occurs, so both lie in $\mathrm{GL}_n(F((z)))$ for F the

Laurent coefficient field, hence are fixed by the turn. Theorem 7 applies to them with $\Gamma = 0$, where Levelt–Turrittin is available: no Hahn field, no product GP , no inequality. The receiver problem was never a property of the comparison.

The pro-Laurent gauge is not the draft’s choice. This corrects the sentence the second version of this memo built its recommendation on. Iritani–Koto transport along the same object: their reconstruction in Section 5.8 uses $M = \bigoplus_j e^{-\varsigma_j^\circ/z} M_B(\varsigma_j^\circ + s_j; Qq^{-c_1(V)/r})$, described in their footnote as a fundamental solution for the summands with respect to $(Q, s_0, \dots, s_{r-1})$ but not for q and z , satisfying $M = \text{id} + O(z^{-1})$ and lying in $\text{End}[z^{-1}]((q^{-1/r}))[[Q, s]]$. Per bulk degree it is a *polynomial* in z^{-1} — which is the draft’s own bound (4.1a), that G_α has no power of z below $-|\alpha|$ — and it is unbounded only after summing over bulk degrees. The sources meet this object and handle it not by an ordered receiver but by Birkhoff factorization, splitting the z -regular factor Φ from the $\text{id} + O(z^{-1})$ factor M' in $(\Phi^\circ)^{-1} \circ M = M' \circ \Phi^{-1}$. Any repair should be measured against that, not against a receiver.

The residual gap

After the two survivals, the blocker is no longer transport across the comparison. It is one statement:

Let T be smooth projective with quantum connection ∇ , and let $\tau^\bullet$ be a bulk parameter in the positive filtration — Iritani’s $\tau^\circ = q^{-1}[Z] + O(q^{-2})$ on the ambient summand, the $O(q^{-1/(c-1)})$ tail of ς_j° together with s_j on a center summand. Show that the framed formal monodromy of $\nabla|_{\tau=\tau^\bullet}$ has the same primitive-sixth multiplicity as $\nabla|_{\tau=0}$.

Everything else in Proposition 4.7 is proved or reduced to a regular gauge, and **lem:divisor-tagging** inherits this gap and nothing else. Section 7 proves this statement whenever the block carrying ν_6 is a rank-two Jordan block that is regular singular after one shearing, which covers the cubic ambient summand. What is still open is the same statement for an arbitrary block, which is what a center-by-center treatment would require.

5 The blockwise route, and what the flatness identities already give

This section carries out the second route far enough to see what it costs. It produces the exact evolution of the leading block operator under bulk motion and proves that multiplicity-one blocks carry no framed monodromy at all; for coalesced blocks it does not reduce the problem, but identifies why the full formal normal form, rather than the compressed pair (U_i, μ_i) , is what has to be controlled. Conventions are Iritani–Koto’s, from their Theorem 5.1: write $U = E \star_\tau$, $C_a = \phi_a \star_\tau$, and

$$\nabla_a = \partial_a + z^{-1}C_a, \quad \nabla_{z\partial_z} = z\partial_z - z^{-1}U + \mu,$$

with μ the constant grading operator. Horizontal sections satisfy $z^2\partial_z Y = (U - z\mu)Y$.

The two identities

Lemma 10 (Flatness). $[U, C_a] = 0$ and $\partial_a U = C_a + [C_a, \mu]$.

Proof. The first is commutativity and associativity of the quantum product. For the second, expand $[\nabla_a, \nabla_z \partial_z] = 0$. The z^{-2} term is $z^{-2}[U, C_a]$ and vanishes by the first identity; the z^{-1} term is $z^{-1}(-\partial_a U + C_a + [C_a, \mu])$, using $z \partial_z(z^{-1} C_a s) = -z^{-1} C_a s + z^{-1} C_a z \partial_z s$ and $\partial_a \mu = 0$. There is no z^0 term. $\square$

So bulk motion is not an arbitrary negative-loop gauge. The first identity says C_a commutes with every spectral projector P_i of U , hence preserves each generalized eigenspace H_i with $U|_{H_i} = u_i + N_i$; the second says the motion of U itself is expressed through C_a and μ alone.

The block frame

Let $T_a = \sum_j (\partial_a P_j) P_j$ be Kato's transport operator, so that $\partial_a P_i = [T_a, P_i]$ and $P_i T_a P_i = 0$. It is independent of z and has no pole, so it moves the blocks without touching the loop coordinate. For an operator A write $A_i = P_i A P_i$ and $A_{ij} = P_i A P_j$, and let $D_a A_i = \partial_a A_i - [T_a, A_i]$ be the covariant derivative in this frame. A short computation using $\partial_a P_i = [T_a, P_i]$ gives the general rule

$$D_a A_i = P_i (\partial_a A) P_i + P_i [A, T_a] P_i. \quad (6.1)$$

Theorem 11 (Block evolution). *Assume the eigenvalues u_j of U are pairwise distinct as scalars, so that each difference $u_i - u_j$ is invertible. Then, unconditionally,*

$$D_a U_i = C_{a,i} + [C_{a,i}, \mu_i], \quad D_a \mu_i = \sum_{j \neq i} (\mu_{ij} X_{ji} - Y_{ij} \mu_{ji}),$$

where $X_{ji} = P_j (\partial_a P_i) P_i$ and $Y_{ij} = P_i (\partial_a P_j) P_j$ are the unique solutions of the Sylvester equations

$$\mathcal{S}_{ji}(X_{ji}) = -(\partial_a U)_{ji}, \quad \mathcal{S}_{ij}(Y_{ij}) = -(\partial_a U)_{ij}, \quad \mathcal{S}_{ji}(X) := (u_j - u_i)X + N_j X - X N_i,$$

with $(\partial_a U)_{ji} = C_{a,j} \mu_{ji} - \mu_{ji} C_{a,i}$ and $(\partial_a U)_{ij} = C_{a,i} \mu_{ij} - \mu_{ij} C_{a,j}$. If both blocks are semisimple, $N_i = N_j = 0$, then $\mathcal{S}_{ji}$ is multiplication by $u_j - u_i$ and the formula collapses to

$$D_a \mu_i = [C_{a,i}, S_i], \quad S_i := \sum_{j \neq i} \frac{\mu_{ij} \mu_{ji}}{u_i - u_j}. \quad (6.2)$$

Proof. For U : by (6.1) and Lemma 10, $D_a U_i = P_i (C_a + [C_a, \mu]) P_i + P_i [U, T_a] P_i$. Since C_a commutes with P_i , the first term is $C_{a,i} + [C_{a,i}, \mu_i]$; since U commutes with P_i , the second is $[U_i, P_i T_a P_i] = 0$. No perturbation formula is used, so this identity holds whatever the nilpotent parts.

For μ : $\partial_a \mu = 0$, so (6.1) gives $D_a \mu_i = P_i [\mu, T_a] P_i$, which is the displayed sum. To evaluate X_{ji} , differentiate $[U, P_i] = 0$ to get $[\partial_a U, P_i] + [U, \partial_a P_i] = 0$ and compress by $P_j(\cdot)P_i$ with $j \neq i$: the first bracket contributes $(\partial_a U)_{ji}$ and the second contributes $U_j X_{ji} - X_{ji} U_i$, which is $\mathcal{S}_{ji}(X_{ji})$. The operator $\mathcal{S}_{ji}$ is invertible because $u_j - u_i$ is invertible and $X \mapsto N_j X - X N_i$ is nilpotent. The same argument with i and j exchanged gives Y_{ij} . When $N_i = N_j = 0$, substituting the two displayed expressions for $\partial_a U$ makes the terms containing $C_{a,j}$ cancel, leaving $\sum_{j \neq i} [C_{a,i}, \mu_{ij} \mu_{ji}] / (u_i - u_j)$. $\square$

Remark 12 (The clean form does not cover the blocks that matter). An earlier version of this memo stated (6.2) under the weaker hypothesis $N_j = 0$ for the *neighbouring* blocks only. That was an error: differentiating $U P_i = (u_i + N_i) P_i$, the off-block part of $\partial_a (N_i P_i)$ contributes $X_{ji} N_i$,

so the term $-XN_i$ cannot be dropped when the block i itself is nonsemisimple. By Corollary 14 the only blocks that can carry ν_6 are the coalesced ones, and in the draft's cubic calculation that block has $J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \neq 0$. So $N_i \neq 0$ is the case of interest, and (6.2) does not apply to it. The cancellation that produced S_i has to be redone with $\mathcal{S}_{ji}^{-1}$ in place of the scalar $1/(u_j - u_i)$; expanding $\mathcal{S}_{ji}^{-1} = \sum_{k \geq 0} (-1)^k (u_j - u_i)^{-k-1} (L_{N_j} - R_{N_i})^k$ shows the clean commutator is only the leading term of that expansion.

Multiplicity-one blocks carry nothing

This is independent of the deformation and settles, in this normalization, the observation the earlier version of this memo only flagged.

Theorem 13 (Simple blocks are invisible). *Let $(\cdot, \cdot)$ be the Poincaré pairing. Then each μ_i is anti-self-adjoint for the restriction of the pairing to H_i , which is nondegenerate; consequently $\text{tr } \mu_i = 0$ and the spectrum of μ_i is symmetric about 0. If u_i has multiplicity one, then $\mu_i = 0$, the block's formal solution is $e^{-u_i/z}$ times a single-valued formal unit, its framed monodromy is the identity, and it contributes no root of unity to ν_6 .*

Proof. Quantum multiplication is Frobenius, $(a \star b, c) = (a, b \star c)$, so U is self-adjoint; distinct generalized eigenspaces of a self-adjoint operator are orthogonal, so each P_i is self-adjoint and the pairing restricts nondegenerately to H_i . The pairing vanishes unless the degrees are complementary, so μ is anti-self-adjoint; hence so is $\mu_i = P_i \mu P_i$ on H_i . An anti-self-adjoint operator has $\text{tr } \mu_i = \text{tr } \mu_i^\dagger = -\text{tr } \mu_i$ and characteristic polynomial even up to sign, giving the trace and symmetry statements. On a one-dimensional space anti-self-adjointness forces $\mu_i = 0$. Formal decoupling from the other eigenspaces by a gauge $g = 1 + zg_1 + \cdots$ can add terms $z c_1 + z^2 c_2 + \cdots$ to the scalar block, but it does not change its z^0 coefficient, since the only z^0 contribution is $-[g_1, U]_{ii} = [U_i, g_{1,ii}] = 0$ in rank one. So the decoupled block is $z^2 \partial_z y = (u_i + z^2 c_2 + \cdots)y$, whose solution is $e^{-u_i/z}$ times a formal unit with no $\log z$ and no fractional power: single-valued under the turn. $\square$

Corollary 14. ν_6 is carried entirely by the blocks of $E \star$ whose eigenvalue has multiplicity greater than one.

This matches the draft's own cubic computation, where K_0 has eigenvalues $\pm 6r, 0, 0$ and the primitive-sixth exponents come from the repeated 0: the $\pm 6r$ blocks are simple, so by Theorem 13 they contribute the identity, and the draft's indicial roots $1/6$ and $5/6$ are $\pm 1/6$ modulo $\mathbf{Z}$, exactly the symmetry about 0 that the theorem forces on a block residue.

What is left, exactly

Proposition 15 (Isomonodromy criterion, semisimple blocks only). *Suppose $N_i = 0$ and $N_j = 0$ for all j , so that the decoupled block connection is $\nabla_i = z \partial_z + A_i(z)$ with $A_i(z) = -z^{-1} U_i + \mu_i$ and (6.2) applies. Then $D_a \nabla_i$ is an infinitesimal gauge transformation by $\Theta_a(z) = -z^{-1} C_{a,i} + \beta$ if and only if*

$$[\beta, U_i] = 0 \quad \text{and} \quad [\beta, \mu_i] = [C_{a,i}, S_i].$$

The z^{-2} and z^{-1} obstructions vanish identically for that choice of leading term; only the z^0 equation remains.

Proof. Infinitesimally, $\delta A_i = [\Theta_a, A_i] - z\partial_z \Theta_a$. With $\Theta_a = z^{-1}\alpha + \beta$ this is $-z^{-2}[\alpha, U_i] + z^{-1}([\alpha, \mu_i] - [\beta, U_i] + \alpha) + [\beta, \mu_i]$. Under the stated hypotheses Theorem 11 gives $D_a \nabla_i = -z^{-1}(C_{a,i} + [C_{a,i}, \mu_i]) + [C_{a,i}, S_i]$. Taking $\alpha = -C_{a,i}$ kills the z^{-2} term because $C_{a,i}$ commutes with U_i , and reduces the z^{-1} equation to $[\beta, U_i] = 0$; the z^0 equation is the second displayed condition. $\square$

Why this is not yet a reduction of the coalesced case

Proposition 15 is stated where it is true, and that is not where the problem is. Two things fail for a coalesced block, and the second is the deeper one.

First, by Remark 12, S_i is not the right object once $N_i \neq 0$: the full Sylvester inverse has to be carried.

Second, and independently, the compressed pair (U_i, μ_i) does not determine the block's exponents. Formal decoupling of a nonsemisimple block from the other eigenspaces produces higher powers of z , and for a nilpotent leading term the Turruttin reduction reaches down into them. The draft's own cubic calculation is the demonstration. Its reduced rank-two block is

$$z^2 \partial_z \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix} = [J_0 + zD_0 + z^2 E_0 + O(z^3)] \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix}, \quad J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix},$$

with E_0 carrying the entry $-8/81$. The ansatz $\tilde{S}_3 = z^\rho(1 + O(z))$, $\tilde{S}_4 = cz^{\rho+1}(1 + O(z))$ gives $\rho = 2c - 19/18$ and $c(\rho + 1) = \frac{19}{18}c - \frac{8}{81}$, hence $\rho^2 + \rho + \frac{5}{36} = 0$ and $\rho = -\frac{1}{6}, -\frac{5}{6}$. The z^2 coefficient is not decoration: setting its $-8/81$ entry to zero leaves $c(\rho + 1) = \frac{19}{18}c$, whose solutions are $\rho = 1/18$ with $c = 5/9$, and $\rho = -19/18$ with $c = 0$ — the classes $\pm 1/18$ modulo $\mathbf{Z}$, which are eighteenth roots of unity and contribute nothing to ν_6 . So a deformation theory that tracks only U_i and $P_i \mu P_i$ cannot see the primitive-sixth exponents at all, and any proposed evolution theorem must reproduce this example. It is the regression test.

6 An unconditional computation: the count for $X \times \mathbf{P}^1$

This section is independent of Hypothesis 4.7H and of every route discussed below. It was originally written inside the refuted endpoint section, which is the wrong place for a surviving theorem, and is promoted here.

The displacement for a trivial bundle is a pure string shift

Theorem 16. *Let $V = \mathcal{O}_X^{\oplus 2}$. Then the Iritani–Koto initial displacement is $\varsigma_j^\circ = r\lambda_j = \pm 2q^{1/2}$ exactly, a class in H^0 , with no tail.*

Two independent proofs, which is what the claim deserves given that an earlier version of this memo asserted the opposite.

First, from their (5.11), $\varsigma_j^\circ = r\lambda_j + [z^{-1}] \log(q^{c_1(V)/(rz)} F_j(1))$. For a trivial bundle $c_1(V) = 0$, so the exponential prefactor is absent, and all Chern roots vanish, so $F_j(1)$ is a scalar series. A power count in the stationary-phase transform then shows it has no negative powers of z : a factor from $\exp(-\varphi_{\geq 3}/z)$ contributes z^{-a} with s -degree at least $3a$, the remaining tail contributes z^{+b} with $b \geq 0$, and Gaussian pairing of total s -degree m contributes $z^{m/2}$, so the net power is at least $a/2 + b \geq 0$. The z^{-1} coefficient is therefore zero. The $O(q^{-1/r})$ that appears in the sources

is a bound on $F_j(1)$, not a nonvanishing statement about ς_j° ; conflating the two is what the earlier version did.

Second, and needing none of that: $\mathbf{P}(\mathcal{O}_X^{\oplus 2}) = X \times \mathbf{P}^1$, and the small quantum cohomology of a product is the tensor product of the factors'. So $E\star = E_X \star \otimes 1 + 1 \otimes E_{\mathbf{P}^1}\star$, and the generalized eigenspace for the cluster at $r\lambda_j$ is $H^*(X) \otimes v_j$ for v_j the j -th eigenline of $E_{\mathbf{P}^1}\star$. The grading operator compresses to $\mu_X + \text{tr}(P_j\mu_{\mathbf{P}^1})$, and those numbers are degree zero, permuted by the deck action $q^{1/r} \mapsto e^{2\pi i/r} q^{1/r}$, hence equal, and they sum to $\text{tr} \mu = 0$, hence vanish. The j -th summand is $z^2 \partial_z = (U_X + r\lambda_j) - z\mu_X$, which is the cubic's quantum connection at the bulk parameter $r\lambda_j \cdot 1 \in H^0$. Since generalized eigenspaces are canonical rather than chosen, this pins $\varsigma_j^\circ = r\lambda_j$ with no appeal to the transform at all.

Corollary 17. $\nu_6(X \times \mathbf{P}^1, 0) = 4$, as a computed fact at a parameter that indisputably exists.

Proof. An H^0 shift changes the connection by $(\tau^0/z) \cdot \text{id}$, which shifts every exponential factor uniformly and changes no block's regular part. This is exact and needs no smallness, which matters because $\pm 2q^{1/2}$ is of unit order. So each summand contributes the cubic's value at the origin, which is 2 by the block computation of Section 7. $\square$

7 The coalesced block is rigid

This section closes the residual gap of Section 4 for a block of the cubic's type. The route is the one Section 5 said was needed: deform the *full* formal normal form rather than the compressed pair. The device that makes it finite is to shear first. After the shearing the block is regular singular, its residue already contains the z^2 coefficient that the regression test showed to be load-bearing, and — this is the point — the bulk connection loses its pole, so the residue moves by conjugation alone.

Setting

Let Λ be the coefficient field of the small connection, containing $r = (3q)^{1/2}$ and its inverse, and let $B = \Lambda[[\tau]]$ be the formal even bulk germ in finitely many even coordinates τ_a . Write the quantum connection in the conventions of Section 5,

$$z\partial_z Y = (z^{-1}U - \mu)Y, \quad \partial_a Y = -z^{-1}C_a Y,$$

with $U = E\star_\tau$, $C_a = \phi_a\star_\tau$ and μ constant, so that Lemma 10 holds: $[U, C_a] = 0$ and $\partial_a U = C_a + [C_a, \mu]$. Everything below is formal in τ and needs no completion of the loop coordinate, no ordered receiver, and no bulk gauge.

There is one standing hypothesis, verified for the cubic at $\tau = 0$:

- (H1) $U(0)$ has an eigenvalue u_0 whose generalized eigenspace H_0 has rank two and on which the nilpotent part is nonzero; every other eigenvalue differs from u_0 by a unit of Λ .

For the cubic, (H1) holds with $u_0 = 0$ and the other eigenvalues $\pm 6r$.

Earlier versions of this memo carried a second standing hypothesis (H2): that after the shearing of Step 4 the block is regular singular at $\tau = 0$, that is $f(0) = 0$ in the notation there. It was verified for the cubic by inspection, because the draft's D_0 is diagonal. It is now a theorem for every rank-two coalesced block of every smooth projective target (Theorem 22), proved from the Frobenius property of quantum multiplication, and it holds pointwise on the germ rather than only at the base point. Nothing below assumes it.

Step 1: decoupling, and the bulk connection stays block diagonal

Lemma 18. *There is a unique gauge $g = I + O(z)$ over B , block-diagonalizing the z -connection with respect to the generalized eigenspaces of U . In the gauged frame the bulk connection is block diagonal as well, and its leading coefficient on H_0 is $C_{a,0} = P_0 C_a P_0$.*

Proof. Existence and uniqueness of g is the splitting lemma at Poincaré rank one, solved order by order by Sylvester equations whose operators are invertible because the eigenvalue differences are units. For the second statement write the gauged pair as $\tilde{\mathcal{A}} = z^{-1}U + O(1)$, block diagonal, and $\tilde{C}_a = z^{-1}\check{C}_a(z)$ with $\check{C}_a = \check{C}^{(0)} + z\check{C}^{(1)} + \dots$, and expand the flatness identity

$$\partial_a \tilde{\mathcal{A}} = -z \partial_z \tilde{C}_a + [\tilde{\mathcal{A}}, \tilde{C}_a]. \quad (8.1)$$

Its off-block part in bidegree (i, j) , $i \neq j$, has zero left side because $\tilde{\mathcal{A}}$ is block diagonal. At order z^{-2} this reads $(u_i - u_j)(\check{C}^{(0)})_{ij} = 0$, and at each successive order the same invertible factor multiplies the next off-block coefficient. So every off-block coefficient vanishes. Since $g = I + O(z)$, the leading term is unchanged. $\square$

Step 2: the exponential factor, and the commutant

Put $u_0(\tau) = \frac{1}{2} \text{tr}(U|_{H_0})$ and $N := U|_{H_0} - u_0 I$, so N is traceless and $N(0) \neq 0$ is nilpotent. Twisting the block by $\exp(u_0/z)$ removes its exponential factor; this changes neither the regular-singular part nor the framed monodromy. The twist replaces the leading bulk coefficient $C_{a,0}$ by $C_{a,0} - (\partial_a u_0)I$.

Lemma 19. $C_{a,0} = p_a I + q_a N$ with $p_a, q_a \in B$, and $p_a = \partial_a u_0$. After the twist the leading bulk coefficient on the block is $q_a N$.

Proof. $[U, C_a] = 0$ gives $[U|_{H_0}, C_{a,0}] = 0$. A 2×2 matrix over a local ring whose reduction is non-scalar is regular, and $U|_{H_0}$ reduces to $u_0(0)I + N(0)$ with $N(0) \neq 0$; the commutant of a regular 2×2 matrix is free of rank two on I and the matrix itself, hence equals $B \cdot I \oplus B \cdot N$. Taking traces in $\partial_a U = C_a + [C_a, \mu]$ on the block gives $2\partial_a u_0 = \text{tr } C_{a,0} = 2p_a$. $\square$

Step 3: the eigenvalue never splits

Theorem 20 (No splitting). $\det N \equiv 0$ on B . Consequently N is nilpotent at every point of the formal germ, the block keeps a single eigenvalue of multiplicity two, and by (H1) it is a single Jordan block.

Proof. Write $d := \det N \in B$; by (H1), $d(0) = 0$. From Theorem 11 the block leading operator evolves by $D_a U_0 = C_{a,0} + [C_{a,0}, \mu_0]$, so by Lemma 19

$$D_a N = D_a U_0 - (\partial_a u_0)I = q_a N + q_a [N, \mu_0].$$

For a traceless 2×2 matrix $\text{adj}(N) = -N$ and $N^2 = -dI$, so

$$\partial_a d = \text{tr}(\text{adj}(N) D_a N) = -q_a \text{tr}(N^2) - q_a \text{tr}(N[N, \mu_0]) = 2q_a d,$$

using $\text{tr}(N[N, \mu_0]) = \text{tr}(N^2 \mu_0) - \text{tr}(N \mu_0 N) = 0$. Now suppose $d \neq 0$ and let $m \geq 1$ be the order of its lowest nonvanishing homogeneous part d_m in τ . The right side of $\partial_a d = 2q_a d$ has order at least m , while the left side has order $m - 1$ with homogeneous part $\partial_a d_m$. Hence $\partial_a d_m = 0$ for every a , so $d_m = 0$, a contradiction. Note d is basis independent, so no choice of block frame enters. $\square$

This is the falsifier of Section 5 answered: the reconstruction displacement cannot split the block that carries ν_6 . It also shows the Kato frame is not needed for this step, since $\det N$ is frame independent.

Step 4: shear

By Theorem 20, $N^2 = 0$ and $N \neq 0$, so $\ker N = \operatorname{im} N$ is a free rank-one submodule; choose a frame (e_1, e_2) of H_0 with $Ne_1 = 0$, $Ne_2 = \nu e_1$, $\nu \in B^\times$, depending formally on τ , and let Γ_a be the resulting frame connection, which has no pole in z . Apply $S = \operatorname{diag}(1, z)$ and write the sheared pair as

$$z\partial_z \hat{Y} = \hat{A}\hat{Y}, \quad \hat{A} = z^{-1}A_{-1} + A_0 + zA_1 + \cdots, \quad \partial_a \hat{Y} = -\hat{C}_a \hat{Y}, \quad \hat{C}_a = z^{-1}K_a + G_a + \cdots.$$

Conjugation by S sends $E_{12} \mapsto zE_{12}$, $E_{21} \mapsto z^{-1}E_{21}$ and fixes diagonal matrices, and $-S^{-1}z\partial_z S = -\operatorname{diag}(0, 1)$. Writing the decoupled block connection before shearing as $z^{-1}N + A'_0 + zA'_1 + \cdots$, one gets

$$A_{-1} = f E_{21}, \quad f := (A'_0)_{21}, \quad A_0 = R := \nu E_{12} + \operatorname{diag}((A'_0)_{11}, (A'_0)_{22} - 1) + (A'_1)_{21} E_{21}. \quad (8.2)$$

On the bulk side, the leading coefficient $q_a N = q_a \nu E_{12}$ shears up to order z^0 , and the only surviving pole comes from the $(2, 1)$ entry of the regular part:

$$K_a = k_a E_{21}, \quad k_a := (\check{C}_{a,0}^{(1)} + \Gamma_a)_{21}. \quad (8.3)$$

There is no z^{-2} term, and this is where Lemma 19 does its work: a $(2, 1)$ entry in $C_{a,0}$ would have produced one, and the commutant forbids it.

For the cubic, (8.2) reproduces the draft's own numbers. With $\nu = 2$, $A'_0 = D_0 = \operatorname{diag}(-19/18, 19/18)$ and $(A'_1)_{21} = -8/81$,

$$f = 0, \quad R = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}, \quad \operatorname{tr} R = -1, \quad \det R = \frac{5}{36},$$

so the characteristic polynomial of R is exactly $\rho^2 + \rho + 5/36$ and its roots are $-1/6$ and $-5/6$. The z^2 coefficient enters through $(A'_1)_{21}$, so the regression test of Section 5 is met by construction rather than by accident: the object being deformed below is R , which sees it.

Step 5: the Poincaré pairing survives the decoupling, and kills f

The shearing of Step 4 produces a pole exactly when $f = (A'_0)_{21} \neq 0$. Earlier versions assumed $f(0) = 0$ as a standing hypothesis, verified for the cubic by inspecting the draft's diagonal D_0 . It is not a hypothesis: it is forced by the Frobenius property of quantum multiplication, at every point of the germ and for every smooth projective target. The only thing needed is that the decoupling gauge of Step 1 preserves the pairing, which is what the following lemma supplies — the step the earlier list of open questions identified as missing.

Let G be the Poincaré pairing on the even part, symmetric and nondegenerate. Quantum multiplication is Frobenius, so $U = E\star$ is G -self-adjoint, and μ is G -anti-self-adjoint. Writing $\mathcal{A}(z) = z^{-1}U - \mu$, those two facts are exactly

$$\mathcal{A}(-z)^T G + G \mathcal{A}(z) = 0. \quad (8.4)$$

Lemma 21 (The decoupling gauge is an isometry). *The unique gauge $g = I + O(z)$ of Lemma 18 satisfies $g(-z)^T G g(z) = G$. Hence the decoupled connection satisfies (8.4) with the same constant pairing, and so does each block with the restriction of G to it, before and after the twist by $\exp(u_0/z)$.*

Proof. Let σ be the involution $\mathcal{A} \mapsto -G^{-1}\mathcal{A}(-z)^T G$ on connection matrices, so that (8.4) reads $\mathcal{A}^\sigma = \mathcal{A}$. A direct computation with the gauge action $g \cdot \mathcal{A} = g^{-1}\mathcal{A}g - g^{-1}z\partial_z g$ gives

$$(g \cdot \mathcal{A})^\sigma = h \cdot (\mathcal{A}^\sigma), \quad h := G^{-1}g(-z)^{-T}G,$$

the two terms matching separately: the conjugation term because $h^{-1} = G^{-1}g(-z)^T G$, and the logarithmic-derivative term because $h^{-1}z\partial_z h = -G^{-1}(z\partial_z g(-z))^T g(-z)^{-T}G$. Take g to be the decoupling gauge and put $\tilde{\mathcal{A}} = g \cdot \mathcal{A}$. The generalized eigenspaces of a G -self-adjoint operator are G -orthogonal, so G is block diagonal and $\tilde{\mathcal{A}}^\sigma$ is block diagonal whenever $\tilde{\mathcal{A}}$ is; and $h = I + O(z)$ because $g = I + O(z)$. So h is a second gauge of the normalized form carrying $\mathcal{A}$ to block-diagonal form, and the uniqueness in Lemma 18 gives $h = g$, which is the claim. Restriction to a block is legitimate because G restricts nondegenerately to each block. The twist by $\exp(u_0/z)$ subtracts the scalar $z^{-1}u_0I$ from the block connection, and (8.4) is insensitive to that because the subtracted term is odd in z and self-adjoint. $\square$

Expanding (8.4) for the twisted decoupled block connection $z^{-1}N + A'_0 + zA'_1 + \dots$ coefficient by coefficient gives the parity rule

$$N^T G_0 = G_0 N, \quad (A'_p)^T G_0 = (-1)^{p+1} G_0 A'_p, \quad (8.5)$$

that is, N and the odd coefficients are G_0 -self-adjoint and the even coefficients are G_0 -anti-self-adjoint. The $p = 0$ case is the tracelessness statement of Section 5, now with the decoupling correction included rather than only for $P_i \mu P_i$.

Theorem 22 (The sheared block is regular singular at every point). *Under (H1), $f = (A'_0)_{21} = 0$ identically on B .*

Proof. By (8.5), N is G_0 -self-adjoint, and $N^2 = 0$ by Theorem 20, so for all x, y one has $(Nx, Ny) = (x, N^2 y) = 0$: the image of N is isotropic. In particular $(e_1, e_1) = 0$, and nondegeneracy of G_0 then forces $(e_1, e_2) \neq 0$. Again by (8.5), A'_0 is G_0 -anti-self-adjoint, and for a symmetric form that gives $(A'_0 x, x) = 0$ for every x . Taking $x = e_1$ and expanding $A'_0 e_1 = \alpha e_1 + f e_2$ gives $f(e_2, e_1) = 0$, hence $f = 0$. $\square$

This is stronger than the deleted hypothesis in two ways. It holds pointwise on the germ, with no differential equation and no lowest-order argument, which is what a statement outside the germ needs; and it holds for every rank-two coalesced block of every smooth projective target, so the cubic's diagonal D_0 was not an accident of that example.

Step 6: no irregularity appears, and the bulk connection is regular

Theorem 23 (The sheared block stays regular singular). *$k_a \equiv 0$ on B . Hence at every point of the formal germ the block is regular singular with residue R , and in the sheared frame the bulk connection has no pole in z .*

Proof. Expand (8.1) for the sheared pair. The coefficient of z^{-2} is $[A_{-1}, K_a] = [fE_{21}, k_a E_{21}] = 0$, which is vacuous. The coefficient of z^{-1} is

$$\partial_a(f)E_{21} = k_a E_{21} + f[E_{21}, G_a] + k_a \operatorname{ad}_R(E_{21}).$$

Writing $R = \begin{pmatrix} \alpha & \nu \\ \gamma & \delta \end{pmatrix}$ one computes $\operatorname{ad}_R(E_{21}) = \begin{pmatrix} \nu & 0 \\ \delta - \alpha & -\nu \end{pmatrix}$, whose diagonal is $\operatorname{diag}(\nu, -\nu)$ and does not involve γ . The left side is off-diagonal, so comparing diagonal entries gives $k_a \nu = -f([E_{21}, G_a])_{11}$, and since ν is a unit, $k_a = f h_a$ for some $h_a \in B$. Comparing $(2, 1)$ entries then gives

$$\partial_a f = f \left(h_a (1 + \delta - \alpha) + ([E_{21}, G_a])_{21} \right),$$

a linear equation $\partial_a f = f w_a$, which is now redundant: Theorem 22 already gives $f \equiv 0$ outright, and $k_a = f h_a$ then gives $k_a \equiv 0$. $\square$

Remark 24 (What the flatness route still buys). The displayed equation $\partial_a f = f w_a$ is kept because it is the only place where the argument would survive the loss of the Frobenius property: with $f(0) = 0$ assumed rather than proved, the lowest-order argument of Theorem 20 applies verbatim and yields the same conclusion. In any setting where the Frobenius input is unavailable, or for a block of a connection that is not the quantum connection of a smooth projective target, that is the route to use, at the cost of carrying the hypothesis.

Step 7: the exponents are constant

Theorem 25 (Rigidity of the formal type). *$\partial_a R = [R, G_a]$ for every bulk direction a . Hence the characteristic polynomial of R has coefficients in Λ , constant in τ ; the formal monodromy of the block is conjugate to $\exp(2\pi i R)$ at every point of the formal germ; and the primitive-sixth multiplicity of the block is constant.*

Proof. By Theorem 23 the sheared pair has $A_{-1} = 0$ and $K_a = 0$. The coefficient of z^0 in (8.1) is $\partial_a A_0 = [A_{-1}, H_a] + [A_0, G_a] + [A_1, K_a]$, which collapses to $\partial_a R = [R, G_a]$. A matrix moving by an infinitesimal conjugation has constant characteristic polynomial: $\partial_a \det(XI - R) = 0$ entrywise. Since $\hat{A} = R + O(z)$, the block is regular singular and its formal monodromy is conjugate to $\exp(2\pi i R)$; shearing changed the exponents by integers only, which ν_6 does not see. $\square$

Corollary 26 (The residual gap, closed at the cubic). *For a smooth cubic threefold, the framed formal monodromy of the zero block has $\nu_6 = 2$ at every point of the formal even bulk germ, in particular at any bulk parameter $\tau^\bullet$ in the positive filtration. The statement of Section 4 therefore holds for the cubic ambient summand.*

Proof. Combine Theorem 25 with the computation after (8.2): the characteristic polynomial is $\rho^2 + \rho + 5/36$ everywhere, with roots $-1/6, -5/6$, whose monodromy eigenvalues $e^{\mp\pi i/3}$ are primitive sixth roots of unity. Since the coefficients of the characteristic polynomial are constants of Λ , evaluation at a topologically nilpotent $\tau^\bullet$ is legitimate and returns the same polynomial; no convergence of a bulk gauge is involved. $\square$

Three remarks

Remark 27 (Why this is not the compressed-pair argument). The deformed object is R of (8.2), not the pair (U_0, μ_0) . The shearing has folded the z^2 coefficient of the unsheared block into the residue, so the failure mode of Section 5 — deleting $-8/81$ and moving the exponents to $\pm 1/18$ — cannot occur here: that deletion changes R itself, hence its characteristic polynomial, and Theorem 25 freezes exactly that polynomial.

Remark 28 (Relation to Cai’s Proposition 6). Cai states the $\pm 1/6$ exponents for the *big* quantum connection and proves it by constructing $M = I + \sum M_n t^n$ with $\partial_{t_i} M = -z^{-1} P_i M$, so that $M^{-1} \nabla_{t_i} M = \partial_{t_i}$ and $M^{-1} \nabla_z M$ is bulk independent. That gauge is precisely the pro-Laurent object of Lemma 4.5 and of Iritani–Koto’s Section 5.8: per bulk degree it is polynomial in z^{-1} , unbounded only after summing bulk degrees. Over $\Lambda((z))[[t]]$, with the bulk kept formal, that argument is sound and proves the same conclusion, because M is fixed by the turn and Theorem 7 applies with the bulk variables as formal parameters. What it does not do by itself is license evaluation at a *specialized* bulk parameter with Novikov coefficients, which is where the receiver discussion of Section 3 came from. The argument above avoids the gauge altogether and produces a τ -constant characteristic polynomial, so specialization is a substitution into constants.

The Serre-decoration route, and why it is not taken

Katzarkov–Kontsevich–Pantev–Yu’s Example 6.21 computes the cubic’s atomic composition and separates the zero atom by its Serre automorphism, concluding irrationality in dimension three. Extending that to one stabilization needs surfaces excluded as well. The route is archived rather than pursued: the decoration it turns on is defined for complex analytic F -bundles, while the A -model F -bundle of a projective variety is non-archimedean analytic, and in that setting the authors defer the Euler pairing and the Serre automorphism themselves to forthcoming work. Its criterion is unnumbered prose, and it mixes formal, non-archimedean analytic and complex analytic statements.

Arbitrary Jordan size: what extends, and the one thing that does not

Earlier versions of this section closed with the expectation that the mechanism — commutant of a regular block kills the would-be z^{-2} bulk term, one shearing makes the block regular singular, flatness then forces the bulk pole to vanish — should extend to a single Jordan block of any size with the shearing $\text{diag}(1, z, \dots, z^{m-1})$, since that is what a general center statement needs. Carrying it out separates cleanly into a part that does extend, a bookkeeping statement, and one step where rank two turns out to be genuinely special.

Proposition 29 (Steps 1 to 3 in size m). *Let H_0 be a generalized eigenspace of $U(0)$ of rank m on which the nilpotent part is regular, the other eigenvalues differing from u_0 by units. Put $u_0 = \frac{1}{m} \text{tr}(U|_{H_0})$ and $N := U|_{H_0} - u_0 I$. Then*

1. $C_{a,0} \in B[N] = B \cdot I \oplus B \cdot N \oplus \dots \oplus B \cdot N^{m-1}$, and the coefficient of I is $\partial_a u_0$; after the twist by $\exp(u_0/z)$ the leading bulk coefficient lies in $N \cdot B[N]$, hence is strictly upper triangular in a cyclic frame, so no entry of it shears down to a pole of the bulk connection;

2. $\text{tr}(N^k) \equiv 0$ on B for every $k \geq 1$. Hence N is nilpotent at every point of the formal germ, and regular there because N^{m-1} has a unit entry, so the block keeps one eigenvalue of multiplicity m and a single Jordan block.

Proof. $[U, C_a] = 0$ gives $[U|_{H_0}, C_{a,0}] = 0$, and the commutant of a regular matrix over a local ring is the free module it generates, which is $B[N]$. Write $C_{a,0} = p_a I + Q_a$ with $Q_a := \sum_{j \geq 1} q_a^{(j)} N^j \in N \cdot B[N]$; this much is the commutant statement alone, and the identification of p_a is deferred to the end.

(2) By Theorem 11, $D_a N = Q_a + [Q_a, \mu_0]$. Put $t_k := \text{tr}(N^k)$, so $t_1 \equiv 0$ by construction. Then

$$\partial_a t_k = k \text{tr}(N^{k-1} D_a N) = k \text{tr}(N^{k-1} Q_a) = k \sum_{j \geq 1} q_a^{(j)} t_{k+j-1},$$

the commutator term dropping because N^{k-1} and Q_a commute and the trace is cyclic: $\text{tr}(N^{k-1} Q_a \mu_0) = \text{tr}(N^{k-1} \mu_0 Q_a)$. By Newton's identities the coefficients of the characteristic polynomial of N are polynomials without constant term in $t_1, \dots, t_m$, so Cayley–Hamilton puts every t_l with $l > m$ in the ideal generated by $t_2, \dots, t_m$. The vector $t = (t_2, \dots, t_m)$ therefore satisfies a linear system $\partial_a t = M_a t$ with M_a over B , and $t(0) = 0$ because $N(0)$ is nilpotent. The lowest-order argument of Theorem 20 applies to the vector unchanged, so $t \equiv 0$, and in characteristic zero the vanishing of all power traces forces the characteristic polynomial of N to be λ^m .

(1) With $\text{tr} N^j \equiv 0$ for every $j \geq 1$ now available, $\text{tr} C_{a,0} = m p_a$, and taking traces in $\partial_a U = C_a + [C_a, \mu]$ on the block gives $m \partial_a u_0 = m p_a$. The twist by $\exp(u_0/z)$ replaces $C_{a,0}$ by $C_{a,0} - (\partial_a u_0) I = Q_a$, which lies in $N \cdot B[N]$ and so has no entry on or below the diagonal. $\square$

The shearing bookkeeping is the next thing to fix, because it decides what has to be proved. Conjugation by $S = \text{diag}(1, z, \dots, z^{m-1})$ multiplies the (i, j) entry by z^{j-i} , so writing the twisted decoupled block connection as $z^{-1}N + A'_0 + zA'_1 + \dots$, the coefficient of z^{-k} in the sheared connection collects the entries of A'_p at sub-diagonal distance $i - j = p + k$, and $z^{-1}N$ shears up to the residue. Hence:

the sheared block is regular singular exactly when A'_p vanishes strictly below its p -th sub-diagonal, for every $p \geq 0$.

For $m = 2$ this is the single condition $f = (A'_0)_{21} = 0$. For $m \geq 3$ it is a nested family, deepest first: $(A'_0)_{m1}$, then $(A'_0)_{m-1,1}$, $(A'_0)_{m,2}$ and $(A'_1)_{m,1}$, and so on.

Proposition 30 (Duality in size m , and its limit). *In the cyclic frame $e_i = N^{m-i} e_m$ the Gram matrix of G_0 is anti-triangular Hankel: (e_i, e_j) depends only on $i + j$ and vanishes for $i + j \leq m$, so $G_0 = J P(N)$ with J the reversal matrix and $P(N) \in B[N]$ invertible. Writing $X^{\text{at}} := J X^T J$ for the reflection in the anti-diagonal, the parity rule (8.5) becomes*

$$(A'_p)^{\text{at}} = (-1)^{p+1} P A'_p P^{-1}.$$

The reflection preserves each sub-diagonal, and conjugation by P acts trivially on the deepest one, so $(A'_p)_{m1} = 0$ for p even and is unconstrained for p odd; on the next sub-diagonal the rule only exchanges the two entries with each other.

Proof. Self-adjointness of N gives $(e_i, e_j) = (N^{m-i}e_m, N^{m-j}e_m) = (e_m, N^{2m-i-j}e_m)$, which depends only on $i+j$ and vanishes when $2m-i-j \geq m$; the anti-diagonal value $(e_m, N^{m-1}e_m)$ is a unit by nondegeneracy. Collecting anti-diagonals gives $G_0 = \sum_{t \geq 0} g_{m+1+t} J N^t = J P(N)$. Substituting into $(A'_p)^T G_0 = (-1)^{p+1} G_0 A'_p$ and multiplying by J on the left gives the displayed rule, using $J^2 = I$. The reflection sends (i, j) to $(m+1-j, m+1-i)$ and so preserves $i-j$; and in $P A'_p P^{-1}$ every correction term is of the form $N^a A'_p N^b$ with $a+b \geq 1$, whose $(m, 1)$ entry is $(A'_p)_{m+a, 1-b}$ and therefore out of range. At distance $m-1$ the rule thus reads $(A'_p)_{m1} = (-1)^{p+1} (A'_p)_{m1}$. $\square$

So duality removes exactly one pole coefficient, the deepest, $z^{-(m-1)}$. In rank two that coefficient is the entire pole, which is why Theorem 22 closes the case; for $m \geq 3$ it is one condition out of a growing family. The expectation that the rank-two mechanism extends as it stands was therefore too optimistic at exactly this step, and the extension needs a different input at the base point.

An earlier version of this memo proposed the grading as that input, on the ground that since μ is the grading operator and $E\star$ raises degree by two, $[\mu, U] = U$, making H_0 for the eigenvalue zero μ -invariant with $[\mu, N] = N$; in a μ -homogeneous cyclic frame an operator of μ -weight p is supported exactly on the p -th sub-diagonal, so “ A'_p has μ -weight p ” would be the regular-singularity condition, with nothing left to check entry by entry.

That route is withdrawn: the identity is false over a Novikov ring. The operator μ implements only the cohomological half of the grading, whereas the homogeneity that makes $E\star$ raise degree by two is with respect to the total grading, which includes $\deg Q^d = 2c_1(X) \cdot d$. On the Q^d -part the correct rule is

$$[\mu, U]_{ij} = (1 - c_1 \cdot d) U_{ij},$$

so $[\mu, U] = U$ holds only in Novikov degree zero. The cubic refutes it on its own matrices: with $\mu = -G_X = \text{diag}(-3/2, -1/2, 1/2, 3/2)$ and $U = K_X$ the small even connection matrix of the draft's (4.9a), the entry $(2, 1)$ gives $[\mu, U]_{21} = 2 = U_{21}$, but the entry $(1, 2)$ gives $[\mu, U]_{12} = -12q \neq U_{12} = 12q$. The drawn consequence fails there too: the zero generalized eigenspace is $H_0 = \text{span}_\Lambda \{P^3 - 6qP, P^2 - 21q \cdot 1\}$, and applying μ to the first generator gives $3qP + \frac{3}{2}P^3$, which is not in H_0 . So the cubic is not the rank-two instance of a μ -grading mechanism, and the appearance that the deleted hypothesis (H2) was explained by one was an artifact of the false identity — (H2) is explained instead, and correctly, by Theorem 22, from the Frobenius property.

Nothing proved in this memo used the identity: Section 7, Steps 1–7, uses only $[U, C_a] = 0$, $\partial_a U = C_a + [C_a, \mu]$, the commutant of a regular 2×2 matrix, and the Frobenius input, so Theorem 25 and Corollary 26 are untouched. What is lost is the proposed route for $m \geq 3$. No replacement input at the base point is currently identified; a correct version would have to work with the total grading, in which Q carries degree, rather than with μ alone.

None of this covers a derogatory block, where the commutant is larger than $B[N]$ and Proposition 29 fails at its first step, nor a semisimple block, where the z^{-1} equation instead reads $(I + \text{ad}_R)(C_{a,0} - p_a I) = 0$ and a resonance $\rho_i - \rho_j = -1$ must be excluded before the same conclusion follows.

8 The one-stabilization theorem, and the one step it still needs

Section 7 closes the residual gap for the cubic block, but the draft's own presentation needs the same statement at every intermediate fourfold of a weak factorization, and that is a statement about arbitrary blocks. This section changes where the bookkeeping happens. In the atom formulation the intermediate fourfolds never appear, because the weak-factorization ledger is an identity in the free abelian group on atoms. That removes the transport problem completely. It introduces one requirement of its own, which the first draft of this section wrongly called standard: the invariant must be well defined on atoms, and an atom is a geometric atomic F -bundle modulo isomorphism *and* modulo change of base point within a connected component of the spectral cover. The second equivalence is where the argument is still open, and Section 8 states exactly why.

The invariant

For a smooth projective T and a point b of the even base of its A -model F -bundle, let $\text{Mon}(T, b)$ be the multiset of eigenvalues of the framed monodromy after Levelt–Turrittin separation, and

$$\nu_6(T, b) := \#\{\zeta \in \text{Mon}(T, b) : \zeta = e^{\pi i/3} \text{ or } e^{-\pi i/3}\},$$

with algebraic multiplicity. The same definitions apply to a summand of the spectral decomposition, in particular to a geometric atomic F -bundle. At a point of the base the coefficient field is $\Lambda((z))$, so Levelt–Turrittin is available and Section 2 applies with $\Gamma = 0$: no receiver enters anywhere in this section.

Bulk local constancy

Theorem 31 (The formal type is constant on the base). *Let (H, ∇) be the A -model F -bundle of a smooth projective T over its base B , and let $b \in B$. Then the framed monodromy at every point of the formal germ of B at b is conjugate to that at b . Consequently ν_6 is locally constant on B , hence constant on each connected component, and likewise on each summand of the spectral decomposition.*

Proof. This is Cai's argument in the proof of his Proposition 6, which deserves to be stated in this generality. Write $\nabla_{t_i} = \partial_{t_i} + z^{-1}P_i$ for the bulk directions, with P_i quantum multiplication by ϕ_i , and solve

$$\partial_{t_i} M = -z^{-1}P_i M, \quad M|_{t=0} = I$$

recursively for $M = I + \sum_n M_n t^n$. The recursion produces $M_n \in \text{End}(H) \otimes \Lambda[z, z^{-1}]$, with z -order bounded below by $-|n|$ at each fixed bulk degree, so $M \in \text{GL}(\text{End}(H) \otimes \Lambda[z, z^{-1}][[t]])$; invertibility is automatic from $M = I + O(t)$. By construction $M^{-1}\nabla_{t_i} M = \partial_{t_i}$, and since the ∇_{t_i} commute with ∇_z , so do their conjugates; hence $M^{-1}\nabla_z M = \partial_z + L$ with $\partial_{t_i} L = 0$, and L is the z -connection at b because $M|_{t=0} = I$.

Let Y be a formal fundamental solution at b , possibly after a ramification $z = w^e$, with $\sigma(Y) = Y M_f$ for the turn σ . Then MY is a fundamental solution at the deformed parameter. Every entry of M is a Laurent series in *integral* powers of z , so $\sigma(M) = M$ by Lemma 3, and

$$\sigma(MY) = M\sigma(Y) = (MY)M_f.$$

The framed monodromy is therefore the same matrix in this solution frame, and well defined up to conjugacy over the constants by Proposition 6. A locally constant integer-valued function on a connected base is constant. $\square$

The gap, stated exactly

Theorem 31 is a statement about the *formal* germ: the deformed connection is gauge equivalent to the one at b over $\Lambda[z, z^{-1}][[t]]$, with t formal. The atom equivalence asks for more. Two geometric atomic F -bundles of the same variety are identified when their spectral lifts lie in the same connected component, and two points of one connected component of a non-archimedean analytic space are not in general related by a chain of formal germs. What is needed is local constancy in the analytic topology, and then connectedness.

The obstruction to upgrading is not bookkeeping. It is the same divergence this memo has been circling from the start. The gauge M of Theorem 31 has z -order bounded below by $-|n|$ in bulk degree n , so it converges when the bulk parameter is topologically nilpotent and not otherwise; at a bulk parameter that is merely nearby in the analytic topology, the sum over bulk degrees has no lower bound in the loop coordinate, and M is not a gauge in any ring where the turn argument runs. So the receiver problem of Sections 2 and 3 is not escaped by moving to atoms; it is relocated to the globalization step.

One route to closing it is visible, and it is Section 7 rather than Cai's gauge. That argument produces the differential identity $\partial_a R = [R, G_a]$ between honest matrix-valued functions on the base, and a function with identically vanishing derivative on a smooth connected space in characteristic zero is locally constant. So the shearing argument globalizes where the gauge argument does not, and what it needs in exchange is its two hypotheses at every point of the component: the block must stay a rank-two Jordan block, and it must stay regular singular after one shearing. Theorem 20 propagates nilpotency and Theorem 23 propagates regularity, but neither excludes the locus where the nilpotent part degenerates to zero, which is where the argument would have to be extended.

Remark 32 (What the gap does and does not touch). It does not touch Theorem 35: the nef-canonical input there is pointwise. It does not touch the computation $\nu_6(\alpha(X)) = 2$ at the hyperplane point, nor its stability under specialization, which is Corollary 26. It touches exactly the claim that these two facts may be compared as statements about one atom.

Remark 33 (Why this is not in tension with the receiver problem). The gauge M is the pro-Laurent object of Section 4, and the whole receiver discussion of Sections 2 and 3 was about *specializing* the bulk parameter to an element with Novikov coefficients, where the sum over bulk degrees loses its lower bound in z . No such specialization occurs here: Theorem 31 compares two points of one base along a formal germ, where the bulk parameters are formal variables and M has Laurent coefficients degree by degree. That is exactly the difference between the two formulations, and it is why the atom route is free of the criterion $ew(\Delta\lambda) < \varepsilon$. Section 7 remains the stronger statement where it applies, since it produces a characteristic polynomial constant in τ and so survives specialization; Theorem 31 is the general one.

The invariant descends to atoms

Proposition 34. ν_6 is invariant under isomorphism of geometric atomic F -bundles. It is well defined on atoms if and only if it is in addition constant on connected components of the spectral cover of each variety, which is open.

Proof. Atoms are geometric atomic F -bundles modulo the equivalence generated by isomorphism of atomic F -bundles and by change of the base point within one connected component of the spectral cover of one variety. An isomorphism carries the connection, hence the Levelt–Turrittin separation and the framed monodromy, so it preserves Mon and ν_6 . The second equivalence is exactly the constancy statement, which Theorem 31 supplies along formal germs only. $\square$

It is worth being precise about why the weaker statement does not suffice. One might hope to compare the set of values of ν_6 over the representatives of a class, which is well defined without any constancy. It does not help: the class one is trying to exclude is precisely a class whose value set would contain both 2, from the cubic representative, and 0, from a surface representative, and a non-constant value set is not by itself absurd. The contradiction needs constancy, not a value set.

No atom in dimension at most two carries the invariant

Theorem 35. *Let α be an atom represented by a smooth projective complex variety of dimension at most two. Then $\nu_6(\alpha) = 0$.*

Proof. Induction over the classification, using the two chemical-formula identities of Katzarkov–Kontsevich–Pantev–Yu’s Proposition 5.22, namely $\text{CFF}(\mathbf{P}(V)) = \text{rank}(V) \text{CFF}(T)$ and $\text{CFF}(\text{Bl}_Z T) = \text{CFF}(T) + (\text{codim } Z - 1) \text{CFF}(Z)$, both of which follow from their Theorems 4.5 and 4.11.

A point has a one-dimensional cohomology with trivial connection, so its atom has trivial framed monodromy and $\nu_6 = 0$.

If K_T is nef, their Lemma 5.24 gives a single atom, and their Claim 6.15 gives a gauge, by u^g with $g = \text{Gr} + \frac{1}{2}T_d$, after which the connection has a first-order pole with nilpotent residue. Undoing that gauge shifts the exponents by the eigenvalues of g , which lie in $\frac{1}{2}\mathbf{Z}$. So the framed monodromy is unipotent on even cohomology and has eigenvalue -1 on odd cohomology, and $\nu_6 = 0$. This input is pointwise and needs no constancy statement: their proof shows that for K_T nef, at any even base point with vanishing H^0 -coordinate, quantum multiplication by the Euler field moves the grading eigenspace for m into eigenspaces for values at least $m + 1$, so its block components κ_{ij} vanish for $j \leq 0$ and the gauged operator has a first-order pole. The normalization of the H^0 -coordinate costs nothing, since translation in the identity direction only twists by a scalar exponential.

Curves: $\mathbf{P}^1 = \mathbf{P}(\mathcal{O}^{\oplus 2})$ over a point, so its atoms are point atoms; a curve of genus at least one has nef canonical class.

Surfaces: every smooth projective surface is obtained from a minimal one by blowing up points, and a point has $\nu_6 = 0$, so by the blowup identity it suffices to treat minimal surfaces. A minimal surface of Kodaira dimension at least zero has nef canonical class. A minimal surface of Kodaira dimension $-\infty$ is $\mathbf{P}^2 = \mathbf{P}(\mathcal{O}^{\oplus 3})$ over a point, or a geometrically ruled surface $\mathbf{P}(V)$ over a curve; in the first case the atoms are point atoms, in the second they are curve atoms.

In every case the atoms are among those already shown to have $\nu_6 = 0$. $\square$

Note that this argument is entirely at the level of atoms as classes, so no bulk point is chosen and no transport is used; the only pointwise input is Claim 6.15, which is a structural statement about a connection with nef canonical class.

The theorem

Theorem 36 (One stabilization). *Assume ν_6 is constant on connected components of the spectral cover, as discussed in Section 8. Let $X \subset \mathbf{P}_{\mathbf{C}}^4$ be a smooth cubic threefold. Then $X \times \mathbf{P}^1$ is irrational.*

Proof. At the hyperplane point of the base of X , the spectral decomposition has two one-dimensional summands for the nonzero eigenvalues of the Euler operator and one summand $\alpha(X)$ for the eigenvalue 0; this is Katzarkov–Kontsevich–Pantev–Yu’s Example 6.21, and it is the draft’s own K_0 with eigenvalues $\pm 6r, 0, 0$. The draft’s computation, which is Cai’s Proposition 6, puts formal solutions with z^ρ , $\rho \equiv \pm 1/6$, in the rank-two block for the eigenvalue 0; equivalently the sheared residue of Section 7 has characteristic polynomial $\rho^2 + \rho + 5/36$. Hence $\nu_6(\alpha(X)) \geq 2$, and it is well defined by Proposition 34.

Since $X \times \mathbf{P}^1 = \mathbf{P}_X(\mathcal{O}_X^{\oplus 2})$, the projective-bundle identity gives $\text{CFF}(X \times \mathbf{P}^1) = 2 \text{CFF}(X)$, so $\alpha(X)$ occurs in the atomic composition of $X \times \mathbf{P}^1$.

Suppose $X \times \mathbf{P}^1$ were rational. By the non-rationality criterion, their Proposition 5.17 in the equivariant form and Proposition 5.30 in the Hodge form, every atom in its atomic composition would be represented by a smooth projective variety of dimension at most $4 - 2 = 2$. Then $\alpha(X)$ would be such an atom, and Theorem 35 would give $\nu_6(\alpha(X)) = 0$, contradicting $\nu_6(\alpha(X)) \geq 2$. $\square$

The invariant does not need to be computed exactly, and the odd cohomology $H^3(X)$ may well lie in the same summand; only the presence of the two primitive-sixth eigenvalues is used, and they cannot be cancelled because ν_6 counts with multiplicity in a free abelian group.

What is imported, and what is proved here

Imported, and not reproved:

- The blowup decomposition of maximal F -bundles, and the projective-bundle decomposition. These are the theorems of Iritani and of Iritani–Koto, packaged as Theorems 4.5 and 4.11; reproving them would be far larger than this theorem, and no presentation should claim independence of them.
- The chemical-formula identities and the non-rationality criterion, which are formal consequences of those two theorems together with weak factorization.
- Claim 6.15 for a variety with nef canonical class, and Lemma 5.24.
- Weak factorization in characteristic zero.

Proved here or in the draft: constancy of the formal type along formal germs (Theorem 31), invariance of ν_6 under isomorphism (Proposition 34), the low-dimensional vanishing (Theorem 35), the cubic block computation, and the specialization-safe rigidity of Section 7.

Open: constancy of ν_6 on connected components, which is Section 8. Hypothesis 4.7H does not appear anywhere in the chain — no framed operator is transported across a comparison isomorphism — so the residual conditionality is of a different kind and a smaller one, but it is conditionality and the theorem should be stated with it until it is removed.

9 The endpoint: what is proved, and the one statement left

The first version of this section claimed the endpoint unconditionally by inverting Iritani’s change of variables. That was refuted, and the refutation is recorded here because the move is tempting and should not be made again. Iritani–Koto’s (5.13) sets $s_j = \varsigma_j - \varsigma_j^\circ$ and states that the invertible change of variables is the one in those *displaced* coordinates, treated as independent formal variables; Iritani’s Lemma 5.15 is a normalization lemma about initial terms, and it is the unnumbered discussion following it that draws the inverse-function conclusion, in the form that $\vartheta \mapsto (\tau(\vartheta), \{\varsigma_j(\vartheta)\})$ is *formally* invertible over $\mathbf{C}((q^{-1/(r-1)}))[[Q]]$ — invertibility of a formal power series map over a specified coefficient ring, with no germ in sight, and Iritani’s Remark 1.5 says explicitly that Ψ is not expected to be analytic in all the variables; and the well-definedness of the pullbacks along the change of variables is argued by Iritani from the String and Divisor equations together with the reduction of $QDM(X)^{\text{La}}$ and $QDM(Z)^{\text{La}}$ to the smaller rings of his Remark 5.6, not from any general substitution principle. Every version of the decomposition identity in the sources is an isomorphism of modules over a formal germ, not a family of pointwise statements. So a parameter at which a prescribed displaced coordinate vanishes need not exist, and “read the identity in either direction” is false.

What replaces it is better, and it splits the endpoint cleanly into a computed half and one remaining statement.

The remaining statement, and why it is small

Weak factorization can be arranged as a roof $X \times \mathbf{P}^1 \leftarrow \cdots \leftarrow V \rightarrow \cdots \rightarrow \mathbf{P}^4$ with every arrow a blowdown. Descending from the roof is then a composition of *forward* substitutions only, so no inverse is evaluated anywhere and the refuted move is not needed. The centre terms vanish by the low-dimensional lemma, giving $\nu_6(X \times \mathbf{P}^1, \widehat{\tau}_L) = \nu_6(\mathbf{P}^4, \rho_R) = 0$ for the two parameters the two descents produce.

Against that stands Corollary 17. So the endpoint reduces to exactly one statement:

$$\nu_6(X \times \mathbf{P}^1, \widehat{\tau}_L) = \nu_6(X \times \mathbf{P}^1, 0), \text{ or merely } \nu_6(X \times \mathbf{P}^1, \widehat{\tau}_L) \geq 1.$$

Three things make this much smaller than anything attempted above. It concerns *one* variety rather than arbitrary intermediate fourfolds. Only a lower bound is needed, so no control of the other blocks is required. And the blocks carrying ν_6 on $X \times \mathbf{P}^1$ are two copies of the cubic’s rank-two Jordan block, which are exactly the hypotheses of Section 7.

The engine to run is therefore Section 7 along the pencil $t\widehat{\tau}_L$, $t \in [0, 1]$, and not Cai’s gauge: Cai’s gauge at $t = 1$ still has z -order unbounded below across Novikov degrees, so it is not a gauge in $\Lambda((z))$ and the receiver criterion returns. Section 7 produces the differential identity $\partial R = [R, G]$ between z -regular objects, so if the pencil’s connection matrices are polynomial in t per slot — which the dimension axiom gives, each insertion of an $H^{\geq 4}$ class costing at least two — the identity holds over $\Lambda[t]$ and $t = 1$ is inside the domain rather than at its edge.

Owed lemmas

1. Per-slot polynomiality in t along the pencil, from the dimension axiom, stated for the accumulated parameter and not only for one step.

2. Nonvanishing of ν along the pencil. Section 7 needs the nilpotent part nonzero, and over $\Lambda[t]$ that is nonvanishing of a polynomial with $\nu(0) = 2$, not an open condition one may assume.
3. Legitimacy of the descent's substitutions past the first step, where the base parameter is evaluated at the accumulated parameter rather than at the origin. The accumulated parameter has only $H^{\geq 4}$ components, so finitely many insertions survive per Novikov degree; this is the same finiteness as item 1 and should be proved once.
4. The low-dimensional lemma, restricted to the parameters that actually arise rather than stated for all of them, since the unrestricted version invokes the operation identity at arbitrary parameters and that is exactly what the refutation forbids.

10 Routes, in the order I would now try them

The framing route has been removed from this list by the checks of Section 4, the blockwise route has been carried to its conclusion for a block of the cubic's type in Section 7, and the endpoint theorem is proved in Section 8. So this list no longer ranks attacks on an open problem; it records what each remaining route would still buy, which is the general statement rather than the endpoint.

1. **Blockwise, avoiding transport altogether.** Developed in Section 5 and completed in Section 7 for a single Jordan block of rank two. What remains of this route is generalization, not repair: a single Jordan block of arbitrary size, by the same shearing with $\text{diag}(1, z, \dots, z^{m-1})$, and the semisimple case with its resonance condition. That generalization is what a statement covering arbitrary centers would need; it is not needed for the cubic endpoint. Signs follow the displayed connection of Iritani–Koto's Theorem 5.1 and should be rechecked against the draft's normalization before anything is written into the manuscript.
2. **Birkhoff.** The identity $(\Phi^\circ)^{-1} \circ M = M' \circ \Phi^{-1}$ of Iritani–Koto's Section 5.8, and its counterpart in Iritani's Section 5.8.2, already factors the bulk transport into a z -regular factor, which Theorem 7 handles, and a factor $M' = \text{id} + O(z^{-1})$. What must be shown is that the second factor does not change the primitive-sixth multiplicity. This is a statement about a $z = \infty$ -side gauge acting on a $z = 0$ invariant, so it is not automatic, and it is close to the original difficulty in another form: any proof will have to use where M' comes from, which is the flat quantum connection — that is, the identities route 1 already exploits directly. Worth keeping as a cross-check rather than as the primary attack.
3. **The low-dimensional centers directly:** primitive-sixth vanishing after Iritani's specialization, using the classification the draft already proves, rather than through a general transport statement.
4. **Divisor tagging on its own terms:** whether the divisor equation gives a more explicit transport than the general positive-bulk gauge of Lemma 4.5. Their Lemma 2.24 is the same mechanism in print and is worth citing; it does not close the lemma, for the reasons under check three.

5. **Normalize, then verify.** Replace “choose L so large” by the smallest weight satisfying separation, adopt the single-copy receiver of Section 3, and verify $ew(\Delta\lambda) < \varepsilon$ case by case. This is fallback machinery now.

The iterated receiver of the archived iterated-receiver discussion is not on the list: it needs a semipositivity hypothesis on the ambient variety that weak factorization does not supply. It remains worth having where it applies. Restricting Proposition 4.7 to centers satisfying the criterion is likewise available, at a cost in generality that is the author’s call.

The observation that multiplicity-one blocks contribute nothing, flagged in the second version of this memo on the strength of Dubrovin’s zero-diagonal fact in canonical coordinates, is no longer a flag: Theorem 13 proves it in this normalization, from self-adjointness of $E^\star$ and anti-self-adjointness of μ for the Poincaré pairing. So every route above needs to control only the coalesced blocks.

A separate scope reduction stands on its own, unchanged: a direct quantum Künneth computation for $X \times \mathbf{P}^1$ together with a direct computation for $\mathbf{P}^4$ would leave the blowup step as the only operation formula the headline theorem needs, with the genus-eight corollary waiting. Note that this cannot be had by triviality of the bundle alone — see the end of check one.

11 Placement in the manuscript

Stated by label because these insertions renumber Section 4. Items 1 to 3 and item 5 are unconditional and can be written now; the checks of Section 4 also make item 6 partly unconditional, since the comparison isomorphisms themselves are z -polynomial and the transport lemma applies to them with no receiver. With Section 7 the step that removes the positive-filtration bulk parameter is unconditional as well for the cubic ambient summand, so what is left conditional is the corresponding step at an arbitrary center. A ninth item is therefore owed: the rigidity theorem itself, as a lemma before `prop:framed-operations`, stated for the block type it covers and with its two hypotheses displayed rather than assumed. Nothing here should be written into the manuscript before the sign conventions are rechecked against the draft’s normalization.

1. The ground field and solution algebra go immediately before “Apply Levelt–Turrittin after a ramification $z = w^e$ ”, which already refers to a universal formal solution algebra that is never constructed.
2. Lemma 4 and Proposition 6 go after `def:framed-sixth-multiplicity`, converting that definition into a characterization, with Remark 5 stating the hypothesis that makes it available.
3. Theorem 7 becomes `lem:frame-transport`, after the paragraph beginning “The frame matters” and before `def:pro-laurent-gauge-group`.
4. In `lem:pv-base-change`, the asserted final clause and the corresponding sentence of its proof become a reference to `lem:frame-transport` — but only once the receiver repair makes that lemma applicable there, since that lemma lives over the pro-Laurent receiver.
5. At (4.1c) in `lem:formal-base-shift`, one sentence identifying M^{bulk} as the transported framed automorphism, in the module frame, with the solution-frame statement cited.

6. `prop:framed-operations` cites the lemma at its three load-bearing steps and says which field plays the role of $\mathcal{H}$ at each. Two of those steps are now unconditional: the transport across Ψ and across Φ needs only that both are z -polynomial with inverses in the same ring, which their statements give, so the field there is $F((z))$ over the Laurent coefficient field and no criterion is verified. The step that removes the positive-filtration bulk parameter is the one that waits.
7. A sentence in `prop:framed-operations` fixing the normalization of the coordinate change, citing Iritani–Koto (5.11)–(5.12) and Iritani’s Section 5.8.1, and noting that any residual constant in a logarithmic Novikov direction is a divisor substitution already covered by (4.1). This is owed regardless of which repair lands, and is check two of Section 4.
8. `lem:divisor-tagging` carries the same obstruction — it embeds the gauge of `lem:formal-base-shift`, whose negative z -order is unbounded, together with every transported operator, into one ordered Hahn field — and needs whatever repair the receiver gets.

Mechanically, a new theorem-like environment must carry exactly one semantic label and a matching row in the claim map consumed by the manuscript’s source-only gate, with an empty declaration list exactly when its coverage is recorded as absent, and the coverage snapshot string in both verification READMEs must be regenerated.

12 Correction trail

The version-by-version correction history has moved to `2026-08-15-c912-frame-transport-memo-archive.tex` together with the superseded routes and the closed checks.

13 Questions

The first question of the second version — whether the two connections compared in Proposition 4.7 lie in the class Theorem 4.34 governs — has been answered no, three times over, in Section 4. Questions 1 and 2 of the fourth version are answered in Section 7: the object to deform is the residue of the sheared block, which sees the z^2 coefficient by construction, and its characteristic polynomial is constant; and the displacement cannot split the block, because $\det N$ satisfies a linear equation with zero initial value. Question 3 is no longer on the critical path, since the coalescence literature is not used. These remain.

1. Does the argument of Section 7 extend to a single Jordan block of arbitrary size, and to a semisimple block once the resonance $\rho_i - \rho_j = -1$ is excluded? This is what the draft’s own presentation needs at every intermediate fourfold. The endpoint no longer waits on it, by Section 8, so this is now a question about how the result is presented and about the general stabilization statement, not about whether the endpoint holds. Partially answered: the commutant and no-splitting steps extend to any Jordan size (Proposition 29), but duality kills only the deepest pole coefficient (Proposition 30), so for $m \geq 3$ the base-point input must come from the grading instead. The precise remaining step is μ -equivariance of the decoupling gauge on the zero block.

2. *Answered, and the hypothesis is gone.* Hypothesis (H2) — that the sheared block is regular singular at the base point — was structural, not computed. What was missing, the same statement for the decoupling correction, is Lemma 21: the decoupling gauge is an isometry of the Poincaré pairing, so the parity rule (8.5) applies to the corrected coefficients, and Theorem 22 then gives the vanishing pointwise on the germ for every smooth projective target.
3. Does the factor $M' = \text{id} + O(z^{-1})$ of the sources' Birkhoff factorization preserve the primitive-sixth multiplicity? Now a cross-check rather than a route.
4. For the low-dimensional centers, can primitive-sixth vanishing be proved pointwise in the coordinates Iritani's specialization produces, given that the nef-canonical case already yields an integral- z gauge to a regular-singular connection and that Theorem 13 disposes of every multiplicity-one block? A pointwise statement there needs no transport at all, which is the shape the endpoint theorem wants.
5. Which presentation should the manuscript take: the operation formulas with ν_6 transported summand by summand, which needs the generalization in item 1, or the atom formulation, in which the intermediate fourfolds never appear because the weak-factorization bookkeeping happens at the level of atoms as objects?